Let your income pay down your Bitcoin loan.

Receive MUSD, keep your operating cash, and automatically direct a chosen share toward your Mezo loan.

1 · PROBLEM

Borrowing against BTC is easy. Paying it back is a chore.

Manual
Owners must remember, size and send every repayment.

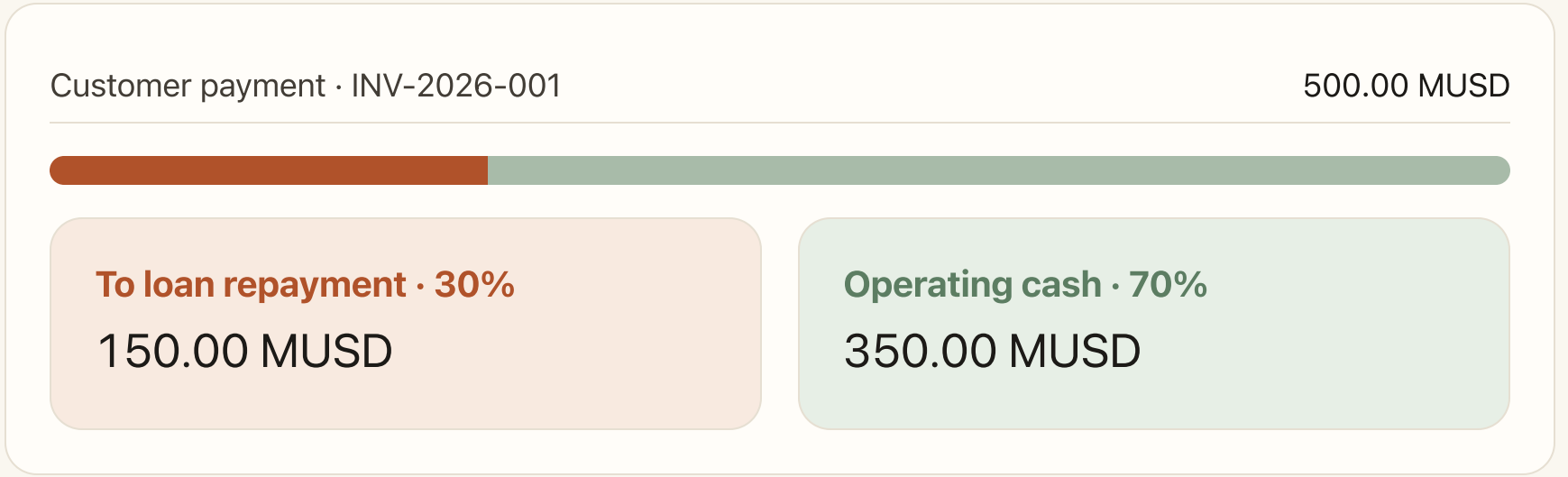
Competing for cash
Revenue lands in the same wallet as spending money.

Disconnected tools
Payment links don't repay loans; loan dashboards don't see revenue.

Who: freelancers, studios and agencies, and BTC-holding founders borrowing MUSD on Mezo.

2 · PRODUCT

Every payment is split once, the moment it arrives.



Real numbers from the live testnet run. Borrowed MUSD is never counted as income.

3 · USER FLOW

Five steps from wallet to automated repayment.

01
Create an account only your wallet controls

02
Account opens and owns a Mezo Trove

03
Set share, reserve, limits, cooldown

04
Share a payment link (approve or permit)

05
Keeper or owner repays; debt and ratio update

The account contract *is* the borrower.

- `BorrowerOperations`: openTrove, repayMUSD, closeTrove, addColl, withdrawColl, claimCollateral (close, addColl, withdrawColl run live)
- `TroveManager.getEntireDebtAndColl`: live debt including accrued interest
- Respects the **1,800 MUSD minimum net debt** and 200 MUSD gas compensation; caps repayment in-block
- MUSD EIP-2612 permit for one-transaction customer payments
- Liquidation, redemption, surplus claim and recovery mode handled

5 · LOAN SAFETY & SECURITY

Guard the loan, close it from income, and nothing else.

Collateral guard

Below the owner's trigger ratio, anyone can call `protect`: pending MUSD repays debt first, then BTC from the owner's buffer goes into the same Trove via `addColl`. Live: **158.2% → 177.4%**.

Auto-close (new in v3)

Partial repayment stops at Mezo's minimum debt. With the owner's opt-in, income keeps saving, and once pending covers the payoff the keeper calls `executeClose`. Live: **1,801.80 MUSD** burned, **0** from operating cash.

Keeper cannot

Borrow, withdraw collateral, touch operating cash, change owner, policy or guard, pay itself, or send funds anywhere but the account's own Trove.

Admin-less factory · atomic clone init · nonReentrant · 89 Foundry tests incl. permission matrix, fuzz & 8 invariants · 3 fork tests on real Mezo bytecode · Sourcify exact match. Self-review, not an audit.

6 · WORKING EVIDENCE (MEZO TESTNET, V2 AND V3)

Real Trove, real payments, real debt reduction.

Account opened its own Trove · 0.045 BTC	debt 2,202.00 MUSD
Customer paid 500 + 100 (permit) + 100 MUSD	30% → repayment
Keeper repaid, capped by owner policy	2,202.00 → 2,102.00
Guard: pending repaid + 0.003 BTC buffer added	158.2% → 177.4%
Keeper auto-close (v3): `executeClose` from pending only	1,801.80 burned · 0.045 BTC back
Owner addColl / withdrawColl · owner closeLoan (v1, v2)	closeTrove
Non-owner setPolicy on-chain	reverted · NotOwner()

Factory v3 0x2AF4eB61EFdF3942B71B3f288FB58Cf719087eeb · app mezo-repay.vercel.app

docs/EVIDENCE.md

7 · REVENUE MODEL & ROADMAP

The business pays a small, capped fee only when the loan gets safer or smaller.

Repayment fee

0.25% of each automated repayment, capped at **5 MUSD**. 150 MUSD repaid → 0.375 MUSD.

Guard fee

0.5% of value moved by `protect`, capped at **10 MUSD**. Auto-close: flat **5 MUSD** per loan closed.

Who pays

The account owner (the business). Payers pay nothing. Testnet fee is zero; fees are planned, not deployed.

Why: 10,000 MUSD/month revenue at 30% → ≈7.50 MUSD/month in fees to keep LTV falling and a guard watching the ratio. Next: real pilot users, redundant keepers, invoice exports, audit before mainnet.